

LangChain Retrievers Complete Guide (Including Similarity Search)

What is a Retriever?

A retriever finds relevant documents from a knowledge base before passing them to the LLM.

Retriever Workflow

Documents → Chunking → Embeddings → Vector DB → Retriever → Relevant Chunks → LLM.

VectorStore Retriever

Uses embeddings and similarity search. Common in RAG systems.

Similarity Search

Compares query embeddings with document embeddings using cosine similarity, dot product, or Euclidean distance.

MMR Retriever

Balances relevance and diversity to avoid duplicate results.

Similarity Score Threshold

Returns documents only above a confidence threshold.

BM25 Retriever

Keyword-based retrieval without embeddings.

Ensemble Retriever

Combines BM25 and vector search for hybrid retrieval.

MultiQuery Retriever

Generates multiple query variations using an LLM.

Parent Document Retriever

Retrieves small chunks but returns larger parent context.

Contextual Compression Retriever

Filters retrieved content to reduce token usage.

Self-Query Retriever

Uses metadata filters automatically.

Time Weighted Retriever

Prioritizes recent documents.

Merger Retriever

Combines outputs from multiple retrievers.

External Retrievers

Wikipedia, Tavily, Arxiv, Amazon Kendra.

Production Stack (2026)

Embeddings → Qdrant/Pinecone → Similarity + BM25 → Ensemble → Reranker → LLM.

Retriever Comparison

- VectorStore: Standard RAG
- BM25: Exact keyword matching
- Ensemble: Hybrid retrieval
- MultiQuery: Better recall
- ParentDocument: Large PDFs
- Compression: Reduce tokens
- SelfQuery: Metadata filtering
- TimeWeighted: Chat memory/news